

- RES.6-012 Introduction to Probability - Lecture 9, 10  
Textbook Notes - Sections 3.5, 3.6, 3.7  
↳ Continuous RVs

## Section 3.5 - Conditioning

### Conditioning on an Event:

The conditional PDF of  $X$  given event  $A$  with  $P(A) > 0$  is defined as

↳ any nonnegative function  $f_{X|A}$  satisfying:

$$P(X \in B|A) = \int_B f_{X|A}(x) dx$$

↳ original PDF restricted to the conditioning event  
& renormalised to integrate to 1

↳ Special case where  $A = \{X \in A\}$ :

$$f_{X|\{X \in A\}}(x) = \begin{cases} \frac{f_X(x)}{P(X \in A)} & ; x \in A \\ 0 & ; \text{otherwise} \end{cases}$$

↳ retains same shape as  $f_X$  inside  $A$

↳ scaled so area equals 1 with 0 outside the area

### Total Probability Theorem for PDFs

If  $A_1, \dots, A_n$  partition the sample space:

$$f_X(x) = \sum_{i=1}^n P(A_i) \cdot f_{X|A_i}(x)$$

↳ Divide & Conquer tool for PDFs: break unconditional distribution into simpler conditional ones, weight each by their probability, sum.

→ Metro Example: Your arrival time  $X$  uniform  $[7:10, 7:30]$

↳ Waiting time depends if catch 7:15 or 7:30 train

↳ Condition each case → simple uniform conditional PDF

↳ TPT gives unconditional PDF of  $Y$

↳ piecewise constant function denser in  $[5:15]$  min range.

- Memoryless Property of the Exponential:  
↳ Extremely important!

$T$  is exponential with parameter  $\lambda$ .

Given  $T > t$  (bulb survived  $t$  units), remaining lifetime  $X = T - t$  has same exponential distribution:

$$P(X > x | T > t) = \frac{P(T > t+x)}{P(T > t)} = \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}$$

original exponential.

- ↳ Residual lifetime is independent of how long process has already run.
- ↳ Exponential  $\rightarrow$  has no memory of the past.
- ↳ Equivalent to memoryless property of Geometric distribution.
- ↳ Defining characteristic making Exponential the model for waiting times in Poisson processes.

## • Conditioning on Another Random Variable

↳ Discrete vs Continuous divergence.

For discrete RVs, conditional PMF:

$$P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

For continuous RVs,  ~~$P(Y=y)$~~   $P(Y=y) = 0$  for specific  $y$ .

↳ Instead: define conditional PDF  $f_{X|Y}(x|y)$  using a limiting argument—condition on  $\{y \leq Y \leq y+\delta\}$  & take  $\delta \rightarrow 0$ :

$$f_{X|Y}(x|y) = \lim_{\delta \rightarrow 0} \frac{f_{X,Y}(x,y)}{f_Y(y)} \text{ for } f_Y(y) > 0.$$

$\rightarrow$  Three-way Relationship:

$$f_{X,Y}(x,y) = f_Y(y) \cdot f_{X|Y}(x|y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f_Y(y) \cdot f_{X|Y}(x|y) dy.$$

## • Conditional Expectation:

$$E[X|A] = \int_{-\infty}^{\infty} x \cdot f_{X|A}(x) dx$$

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y) dx$$

→ Expected Value Rule:

$$E[g(X)|Y=y] = \int_{-\infty}^{\infty} g(x) \cdot f_{X|Y}(x|y) dx$$

→ Total Expectation Theorem → two forms:

$$E[X] = \sum_{i=1}^n P(A_i) \cdot E[X|A_i]$$

$$E[X] = \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y) dy.$$

↳ unconditional expectation equals the weighted average of conditional expectations.

Piecewise PDF Example:  $f_X$  is  $\frac{1}{3}$  on  $[0,1]$ ;  $\frac{2}{3}$  on  $(1,2]$ .

↳ Define  $A_1 = \{X \in [0,1]\}$ ,  $A_2 = \{X \in (1,2]\}$  with  $P(A_1) = \frac{1}{3}$  &  $P(A_2) = \frac{2}{3}$

↳ Conditional PDFs are uniform on each piece.

↳ Easier to compute  $E[X|A_i]$  for each piece & apply TET, than it is to integrate the piecewise PDF directly.

## • Independence of Continuous RVs:

$X$  and  $Y$  are independent if their joint PDF factorises:

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x,y.$$

↳ Equivalently:  $f_{X|Y}(x|y) = f_X(x)$  → knowing  $Y$  provides no info about  $X$ .

↳ Results of Independence:

$$E[g(X)h(Y)] = E[g(X)] \cdot E[h(Y)]$$

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$$

## • Section 3.6 - Continuous Bayes Rule

### • The Inference Problem :

→ The Setting: an unobserved quantity  $X$  has a known prior PDF  $f_X$ .  
You make a noisy measurement  $Y$ , modelled by a conditional PDF  $f_{Y|X}(y|x)$ .

Given the observed value  $y$ , what can you infer about  $X$ ?

↳ Answer : the posterior PDF  $f_{X|Y}(x|y)$ .

↳ Inference is a central problem of statistics.

### • Bayes' Rules - Continuous Version

From the relationship  $f_{X,Y}(x,y) = f_X(x) \cdot f_{Y|X}(y|x) = f_Y(y) \cdot f_{X|Y}(x|y)$  :

$$f_{X|Y}(x|y) = \frac{f_X(x) \cdot f_{Y|X}(y|x)}{f_Y(y)}$$

↳ Denominator - determined from TPT

$$f_Y(y) = \int_{-\infty}^{\infty} f_X(t) \cdot f_{Y|X}(y|t) dt$$

↳ Formula :

- $f_X(x)$  - prior : what you believe about  $X$  before observing  $Y$ .
- $f_{Y|X}(y|x)$  - likelihood : how probable the observation  $y$  is if  $X=x$ .
- $f_{X|Y}(x|y)$  - posterior : updated belief of  $X$  after observing  $Y$ .
- $f_Y(y)$  - normalising constant : ensures posterior integrates to 1.

↳ {Prior → Likelihood → Posterior} structure used in Bayesian Inference

## • Mixed Case - Discrete Unknown, Continuous Observation

In many practical problems the unknown is discrete but the measurement is continuous.

A binary signal  $S \in \{-1, +1\}$  observed through Normal Noise is an example.

Bayes Rule Adapts: 
$$P(N=n | Y=y) = \frac{P_N(n) \cdot f_{Y|N}(y|n)}{\sum_i P_N(i) \cdot f_{Y|N}(y|i)}$$

↳ Denominator mixes a sum (over discrete values of  $N$ ) with a conditional PDF (continuous observation).

Signal Detection Example:

↳ Binary Signal  $S \in \{-1, +1\}$  with  $P(S=1) = p$

↳ Received signal  $Y = S + N$  where  $N$  is standard Normal.

→ Given observation  $y$ :

$$P(S=1 | Y=y) = \frac{p \cdot e^{-(y-1)^2/2}}{p \cdot e^{-(y-1)^2/2} + (1-p) \cdot e^{-(y+1)^2/2}}$$

↳ Probability increases monotonically in  $y$ .

↳ Large positive observation strongly suggest  $S=1$  was transmitted.

↳ Formula shows how observations changes belief about which signal was sent.

## • Reverse Direction - Continuous Unknown, Discrete Observation

↳ Use Bayes Rule to infer continuous  $Y$  from discrete observation  $A$ .

$$f_{Y|A}(y) = \frac{f_Y(y) \cdot P(A | Y=y)}{\int_{-\infty}^{\infty} f_Y(t) \cdot P(A | Y=t) dt}$$

↳ All forms of Bayes → continuous/continuous; continuous/discrete; discrete/continuous.

↳ have the same structure

↳ Numerator = Prior × Likelihood

↳ Denominator = Normalising constant to make result integrate/sum to 1.

## • Section 3.7 - Summary

### • Chapter 2 & 3 → Parallel Structure

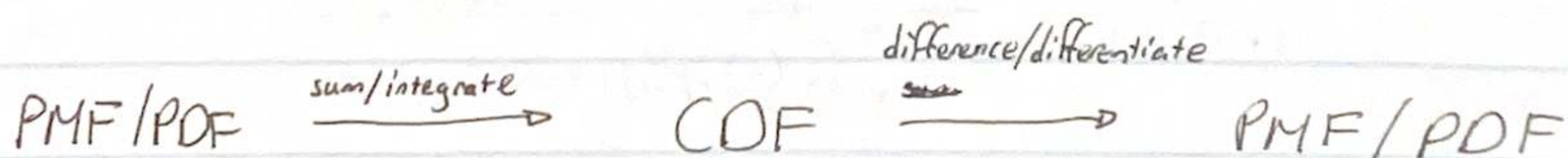
↳ direct counterpart for every discrete/continuous concept.

| Concept                 | Discrete (Ch. 2)      | Continuous (Ch. 3)       |
|-------------------------|-----------------------|--------------------------|
| Probability Law         | PMF $P_X(k)$          | PDF $f_X(x)$             |
| Probability Calculation | Sum                   | Integral                 |
| Expectation             | $\sum x \cdot P_X(x)$ | $\int x \cdot f_X(x) dx$ |
| Conditioning            | Conditional PMF       | Conditional PDF          |
| Total Probability       | Sum over partition    | Integral over $f_Y$      |
| Bayes                   | Discrete formula      | Continuous formula       |
| Unifier                 | CDF                   | CDF                      |

### • Named Distributions

| Distribution             | PDF                                                           | Mean            | Variance             |
|--------------------------|---------------------------------------------------------------|-----------------|----------------------|
| Uniform $[a, b]$         | $\frac{1}{b-a}$ on $[a, b]$                                   | $\frac{a+b}{2}$ | $\frac{(b-a)^2}{12}$ |
| Exponential $(\lambda)$  | $\lambda e^{-\lambda x}, x > 0$                               | $1/\lambda$     | $1/\lambda^2$        |
| Normal $(\mu, \sigma^2)$ | $\frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-(x-\mu)^2/2\sigma^2}$ | $\mu$           | $\sigma^2$           |

### • CDF:



↳ Whenever a direct PMF or PDF calculation is difficult

↳ Maximum of several RVs, functions of RVs, etc.

↳ Strategy: Compute CDF first and then use it to recover the PMF/PDF